



OPEN Anger and emotion regulation strategies: a meta-analysis

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The hypothesis that maladaptive emotion regulation plays a central role in anger problems has driven the development of cognitive-behavioral interventions for anger and has stimulated a wealth of studies. However, this work is heterogeneous, drawing on multiple theories from clinical psychology and affective science, and focusing on multiple dimensions of emotion regulation. In addition, previous findings have not always been consistent. The present meta-analysis aimed to characterize associations between anger and emotion regulation strategies (i.e., avoidance, acceptance, distraction, reappraisal, rumination, suppression). A systematic search was conducted, and 81 studies (115 effect sizes) were included in the analysis. The results indicated consistent positive associations between anger and avoidance, rumination, and suppression, and consistent negative associations between anger and acceptance, and reappraisal. The relation between anger and distraction could not be analyzed. Heterogeneity was large in all analyses, and multiple differences between studies, including type of anger, clinical status, criminal status, type of culture, study quality, and sample sex distribution, influenced the effect sizes. Overall, the present findings support the view that anger is consistently associated with the differential use of multiple emotion regulation strategies and suggest methodological improvements for future studies.

Keywords Anger, Emotion regulation, Psychopathology, Meta-analysis

Anger is a common emotional experience in everyday life^{1,2}. It unfolds when individuals believe that they will not attain a goal due to the actions of another person, and that the situation can be changed³. The putative function of anger is to promote confrontation with others, which can be adaptive in certain contexts⁴, but it can also escalate to aggressive behavior^{5,6}. When anger is too frequent or too intense, it is associated with a range of antisocial behaviors including domestic violence⁷, childhood maltreatment⁸, aggressive driving⁹ and even homicide¹⁰. Furthermore, increased anger has been documented in multiple mental disorders, being associated with greater symptom severity, greater comorbidity and lower quality of life^{11–13}.

According to an influential perspective, maladaptive emotion regulation is a central mechanism underlying problematic anger and its negative implications^{14,15}. This view has driven the development of cognitive-behavioral interventions in anger^{16–18} and has stimulated a wealth of studies¹⁵. However, this work is extremely heterogeneous, drawing on multiple theories from clinical psychology and affective science, and focusing on multiple dimensions of emotion regulation.

Theoretical perspectives on anger and emotion regulation

One of the theoretical frameworks that have focused on the role of emotion regulation in anger and aggression has been articulated by Robertson and coworkers¹⁵. They argue that aggression is driven by maladaptive regulation of negative affect and particularly anger. According to this view, there is a continuum of maladaptive emotion regulation “styles” underlying aggression, ranging from emotion under-regulation to emotion over-regulation. Emotion under-regulation refers to failures to regulate negative affect (e.g., anger) in a sufficient manner so as to maintain goal-directed behavior. Over-regulation relies on attempts to block negative affect altogether, either by avoiding unpleasant emotional experiences or the situations that may trigger it, or by suppressing its expression. These maladaptive emotion regulation styles involve a variety of strategies, which are not inherently maladaptive, but become so when used improperly (e.g., irrespective of situational demands), making them inefficient, or when employed to avoid unpleasant emotional experiences¹⁵.

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Anger avoidance is also in the focus of Gardner and Moore's theoretical framework¹⁴. Rumination, an emotion regulation strategy that involves repetitive thinking about negative emotions¹⁹, is viewed here as a cognitive form of anger avoidance, whereas aggressive behavior is conceptualized as a behavioral form of anger avoidance. When used to escape anger, both strategies are hypothesized to be maintained through negative reinforcement¹⁴.

Emotion regulation is an important part of the integrative cognitive model of trait anger²⁰. This theory argues that certain individuals are prone to automatic hostile interpretations and consequently, more frequent anger. Rumination, which is conceptualized in this model as a form of attentional dysregulation, can further contribute to the amplification of anger. In contrast, controlled processes can buffer anger reactions, through the reappraisal of the situation as non-hostile (for a similar role in aggressive behavior, see ²¹) and through the effortful interruption of rumination and distraction from hostile thoughts. Effortful control can also involve suppressing anger and aggressive behavior.²⁰

Overall, these theoretical models assume that maladaptive emotion regulation is central to problematic anger. They also attempt to identify the ways in which emotion regulation can be maladaptive (e.g., under- or over-regulation), and illustrate the variety of strategies involved in both the upregulation (e.g., rumination, suppression) and the downregulation (e.g., reappraisal, distraction) of anger.

Multiple emotion regulation strategies in anger

The focus on multiple emotion regulation strategies in current research dates back to pioneering clinical interventions and basic studies on anger. For example, early work on cognitive-behavioral interventions²² has highlighted the special role of reappraisal in anger regulation. Reappraisal is an emotion regulation strategy that involves reinterpreting the situation in order to modulate its emotional impact²³, and is a central skill targeted by Novaco's stress inoculation training in anger²² and other cognitive-behavioral interventions^{17,18}. The efficiency of these interventions has been consistently supported^{17,18,24}, well before the impact of using reappraisal on reducing anger was corroborated by experimental²⁵ and individual differences²⁶ (e.g., dispositional traits) research. In a similar manner, experimental studies²⁷ have investigated the potential effect of acceptance (i.e., leaving emotion unfold naturally) in reducing anger, in light of the central role attributed to this emotion regulation strategy in acceptance- and mindfulness-based interventions^{28,29}.

Another emotion regulation strategy linked to anger in early work is expressive suppression. This strategy involves inhibiting the expression of emotion, and was found in early experimental work to increase cardiovascular activity (e.g., heart rate, systolic blood pressure) in anger³⁰. Coined as "anger-in", anger suppression has also been assessed as an individual difference^{31–33} and has been associated with a host of emotional and physiological characteristics³⁴.

In studies examining anger, rumination was associated with both increased aggressive behavior and slower physiological recovery (i.e., prolonged cardiovascular reactivity) following anger-inducing events^{35,36}. Notably, rumination was compared either with distraction³⁵, following in the footsteps of early studies on the response styles theory in depression³⁷, or reappraisal²⁵. Similarly, other studies compared between reappraisal, acceptance and suppression, and found that former strategy was more efficient in reducing anger compared to the latter two strategies²⁷.

The present meta-analysis

Robertson et al. (p. 79)¹⁵ have argued that the concept of emotion regulation is clearly cornerstone in current research on problematic anger, but that "unless a cohesive stream of research is established, it may become difficult to draw conclusive findings from different studies." The present meta-analysis set out to offer an integrative perspective on current evidence, by identifying consistent associations between anger and multiple emotion regulation strategies. We focused on six emotion regulation strategies and, in light of current perspectives on the relative efficiency of emotion regulation strategies³⁸, we expected that some (e.g., acceptance, distraction, reappraisal) showed a negative and others (e.g., avoidance, rumination, suppression) showed a positive association with anger.

To reduce heterogeneity related to methodological differences between studies in adults vs. other age categories (e.g., children), we restricted the present meta-analysis to adults. Also, we focused on subjective measures of anger. Another element of our approach, designed to limit heterogeneity, was our focus on the habitual and spontaneous use of emotion regulation strategies. This focus was reflected in self-report questionnaires that assessed both the global use of these strategies and their spontaneous use in specific contexts, such as through ecological momentary assessment. Because these measures did not overlap with the instructed use of emotion regulation³⁹, we decided to exclude experimental studies in which emotion regulation strategies were manipulated.

Another aim of the present meta-analysis was to uncover differences between studies that could explain heterogeneity in previous results on anger and emotion regulation. First, we compared effect sizes in subgroups of studies that focused on trait vs. state anger. Considering that trait anger is a relatively stable individual difference³³, whereas state anger varies across contexts (which may or may not be specified), we expected the former to be more consistently associated with emotion regulation.

Clinical status was also examined as potential moderator considering that problematic anger has been associated with multiple mental disorders⁴⁰. Here, we also considered the distinction between studies that focused on clinical samples (i.e., patients diagnosed with a mental disorder) and those that selected individuals for high levels of anger, involving clinically relevant behavioral dysfunctions⁴¹. We expected higher effect sizes of the associations between anger and emotion regulation in studies on clinical samples and samples with clinically relevant levels of anger, compared to general samples.

Another potential moderator that we considered was criminal status, in light of studies showing that both problematic anger⁴², and maladaptive emotion regulation⁴³ are increased in offenders compared to non-offenders.

Cultural differences were also examined as a potential moderator. Individualist cultures emphasize the role of emotions in individual wellbeing, whereas collectivist cultures prioritize social convention⁴⁴, and these differences influence the use of emotion regulation and its consequences. For instance, in individualist⁴⁵, but not in collectivist cultures⁴⁶, suppression is associated with increased anger. In cultures where anger expression is incongruent with social norms, rumination might also be increased⁴⁷. Therefore, we contrasted subgroups of studies in individualist vs. collectivist cultures.

Sample differences were also investigated as potential moderators, particularly sex distribution and age category. Previous studies on sex differences in the use of emotion regulation strategies such as suppression^{48,49} and rumination^{50,51}, are inconsistent, but this issue has remained open. Previous work has also investigated potential differences in anger regulation between younger and older adults, also with apparently inconsistent results on strategies such as suppression^{52,53} and reappraisal^{54,55}.

Finally, we considered the possibility that methodological differences explained heterogeneity in previous studies. Study design (i.e., cross-sectional vs. longitudinal) and study quality were examined, with the expectation that longitudinal and higher-quality studies showed more consistent associations between anger and emotion regulation.

Results

Study selection and inclusion

As illustrated in Fig. 1, the search returned 9262 articles. After removing 4256 duplicates, a total of 5006 articles were screened based on title and abstract, and 4808 articles were excluded for not meeting the inclusion criteria. The remaining 198 articles were evaluated full-text, and 114 were excluded based on the inclusion and exclusion criteria.

A total of 73 studies met the eligibility criteria, and an additional 11 were also eligible, but did not provide enough data for effect size estimation. The authors of studies in the latter category were contacted and they provided data from 8 studies. Information on missing group characteristics was also requested for 14 of the 82 studies, and 3 authors responded.

Characteristics of included studies

Table 1 shows the characteristics of studies. Eighty-one studies were included in the analysis and provided 115 effect sizes on the association between anger and one of the emotion regulation strategies: suppression ($k=41$), rumination ($k=37$), reappraisal ($k=12$), acceptance ($k=12$), and avoidance ($k=12$). A single study reported data on the association between anger and distraction and was discarded from analyses. Most of these studies (93.82%) were cross-sectional, and the rest ($k=5$) were longitudinal. Thirteen studies reported data from clinical samples, and the rest involved healthy participants.

Global effects

As shown in Table 2, anger was positively associated with avoidance, rumination, and suppression, and negatively associated with acceptance, and reappraisal. Heterogeneity was significant and high in all analyses.

Study quality

The methodological quality of most studies ($k=56$) was assessed as “poor” (see Supplementary Tables 1–2). Twenty-three studies were rated “fair”, and only 2 were rated “good”. Most studies did not provide sample size justification, and did not control relevant confounding variables. Another common limitation was the cross-sectional design.

Subgroup analyses

Table 3 shows the results of all subgroup analyses.

Type of anger (trait anger vs. state anger related to a specific event vs. state anger not related to a specific event) was a significant moderator of the association between anger and suppression, with a medium positive effect size for trait anger, a low positive effect size for state anger related to a specific event, and a non-significant effect size for state anger not related to a specific event. Type of anger was not a significant moderator of the associations between anger and reappraisal, and rumination, and could not be tested as a moderator for the associations with acceptance and avoidance due to the insufficient number of studies per subgroup (i.e., $k < 3$).

Clinical status (clinical diagnosis vs. clinical anger vs. non-clinical) was a significant moderator of the association between anger and suppression, with a high positive effect size in samples with clinical anger, a medium positive effect size in samples with a clinical diagnosis, and a small-to-medium positive effect size in non-clinical samples. The positive effect size of the association between anger and avoidance was also larger in clinical compared to non-clinical samples (clinical anger could not be included in this analysis due to $k < 3$). Clinical status could not be tested as a moderator for the associations between anger and acceptance, reappraisal, and rumination because of the insufficient number of studies per subgroup (i.e., $k < 3$ in more than 1 subgroup).

Criminal status (offenders vs. non-offenders) was a significant moderator in the relation between anger and acceptance, with a larger negative effect size in offenders than in non-offenders. The subgroup analysis could not be conducted for the rest of the associations given that there were fewer than necessary (i.e., $k < 3$) studies per subgroup.

Type of culture (individualist vs. collectivist) was a significant moderator of the relation between anger and rumination, with a larger positive effect size in samples from collectivist cultures, compared to samples from

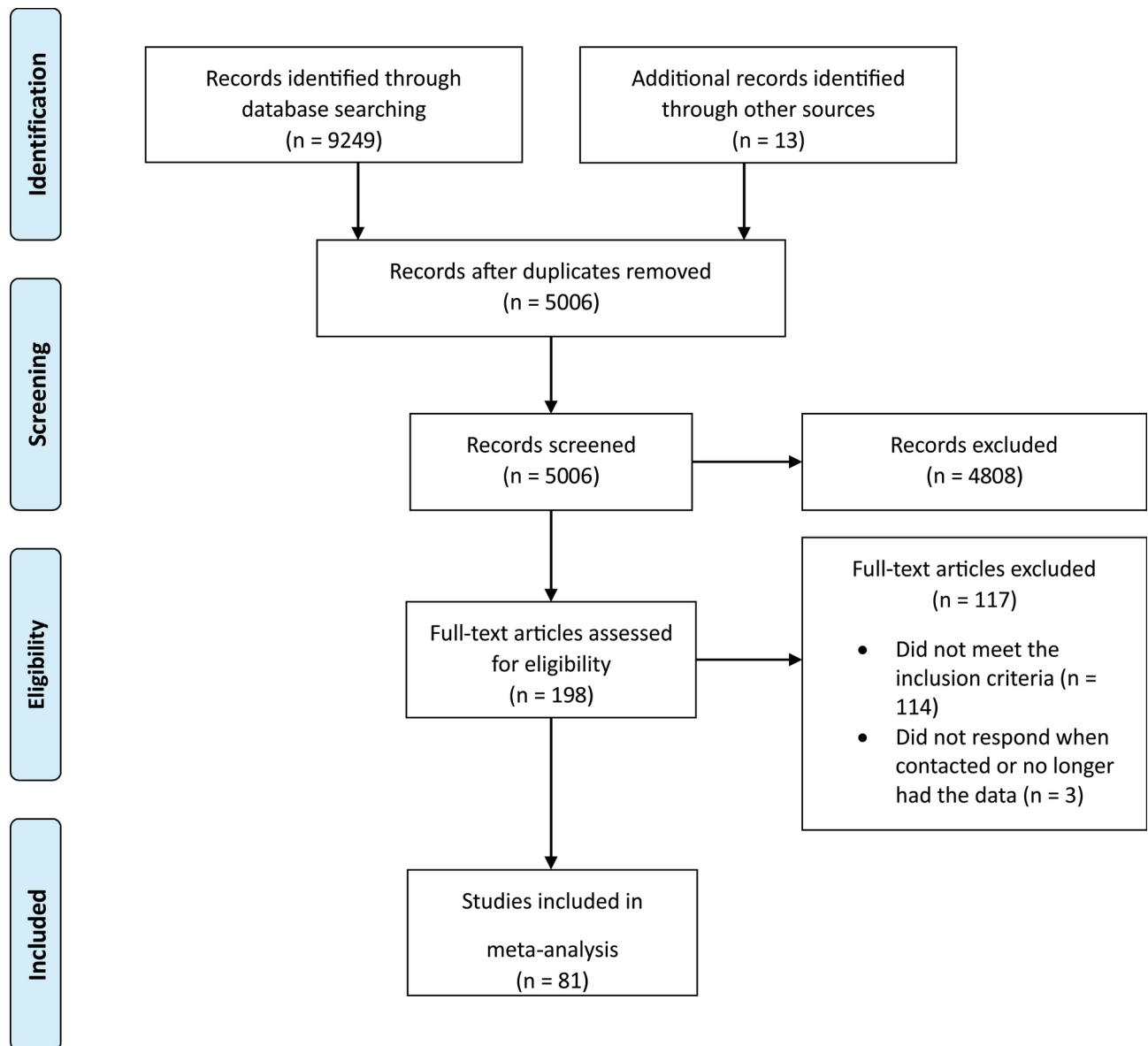


Fig. 1. PRISMA flowchart showing the selection of studies.

individualist cultures. It was also a significant moderator of the relation between anger and suppression, with samples from individualist cultures showing higher positive effect sizes compared to samples from collectivist cultures. The type of culture was not a significant moderator of the relations between anger and avoidance, and anger and reappraisal. The subgroup analysis could not be tested in the relation between anger and acceptance given that all study samples were from individualist cultures.

Type of design (cross-sectional vs. longitudinal) could not be tested as moderator given that there were fewer than necessary (i.e., $k < 3$) studies per subgroup.

Study quality (fair vs. poor; good-quality studies were not included because $k < 3$) was a significant moderator in the associations between anger and acceptance, avoidance, rumination, and suppression. In all subgroup analyses, fair-quality studies showed a higher effect size compared to poor-quality studies. The analysis could not be run on reappraisal given that there were too few studies (i.e., $k < 3$) in one subgroup.

Meta-regressions

Sex distribution. As shown in Table 4, the percent of women was a significant positive predictor of the effect size of the association between anger and acceptance. The association between percent of women and the effect size was not significant in the relations between anger and avoidance, reappraisal, rumination, and suppression.

Age. Mean age was not a significant moderator of the relations between anger and any of the emotion regulation strategies (Table 4).

Study reference	N	Mean age	% women	Country	Individualist/Collectivist	Clinical status	Criminal status	Type of anger	Anger scale	Type of ER strategy	ER strategy scale	Effect size (r)
Akinci and Atintas ⁵⁶	496	32.8	72.4	Turkey	collectivist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Avoidance	CoSS/Avoidance	0.11
Alcázar, et al. ⁵⁷	95	21.67	69.56	Mexico	collectivist	clinical anger	non-offenders	Trait	ML-STAXI/Trait ANG	Suppression	ML-STAXI/ANG IN	0.50
Alcázar-Olán, et al. ⁵⁸	376	20.48	60.9	Mexico	collectivist	clinical anger	non-offenders	Trait	ML-STAXI/Trait ANG	Suppression	ML-STAXI/ANG IN	0.53
Anestis, et al. ⁵⁹	200	18.32	68.5	USA	individualist	non-clinic	non-offenders	Trait	BPAQ/ANG	Rumination	ARS	0.40
Arslan ⁶⁰	468	19.75	55.12	Turkey	collectivist	non-clinic	non-offenders	Trait	STAXI/Trait ANG	Avoidance	CWSS/Avoidance	-0.01
Balsamo ⁶¹	353	32.6	65.2	Italy	individualist	non-clinic	non-offenders	Trait	STAXI/Trait ANG	Rumination	Padua Inventory/Tendency to Doubt and to Ruminare	0.48
Berry, et al. ⁶²	233	19.2	66.5	USA	individualist	non-clinic	non-offenders	Trait	STAXI/Trait ANG	Rumination	DRS	0.60
Besharat ⁶³	933	22.5	58.41	Iran	collectivist	non-clinic	non-offenders	Trait	Tehran MAS/Trait ANG	Rumination	ARS	0.59
Besharat ⁶³	933	22.5	58.41	Iran	collectivist	non-clinic	non-offenders	State	Tehran MAS/State ANG	Rumination	ARS	0.55
Boersma-van Dam, et al. ⁶⁴	111	43.8	80.2	Belgium + the Netherlands	individualist	non-clinic	non-offenders	Event-related State	1-item Anger	Rumination	CERQ/Rumination	0.22
Borders, et al. ¹⁶¹ (Study 1)	170	19.7	66	USA	individualist	non-clinic	non-offenders	Trait	BPAQ/Anger	Rumination	RRQ	0.40
Borders, et al. ¹⁶¹ (Study 2)	134	31.8	60	mixed	individualist	non-clinic	non-offenders	Trait	BPAQ/Anger	Rumination	RRQ	0.47
Borders and Liang ⁶⁵ (Sample 1)	200	-	-	USA	individualist	non-clinic	non-offenders	Trait	BPAQ/Anger	Rumination	ARS	0.38
Borders and Liang ⁶⁵ (Sample 2)	150	-	-	USA	individualist	non-clinic	non-offenders	Trait	BPAQ/Anger	Rumination	ARS	0.47
Borders and Lu ⁶⁶	128	19.48	78	USA	individualist	non-clinic	non-offenders	Trait	BPAQ/Anger	Rumination	ARS	0.45
Borders and Wiley ⁶⁷ (Sample 1)	155	22.49	46.7	USA	individualist	non-clinic	non-offenders	Event-related State	Anger 7-item 7-point Likert	Rumination	RIOS-Modified	0.57
Borders and Wiley ⁶⁷ (Sample 2)	464	19.56	78	USA	individualist	non-clinic	non-offenders	Event-related State	Anger 7-item 7-point Likert	Rumination	RIOS-Modified	0.28
Borders and Wiley ⁶⁷ (Sample 3)	211	19.45	100	USA	individualist	non-clinic	non-offenders	Event-related State	Anger 7-item 7-point Likert	Rumination	RIOS-Modified	0.43
Brassard, et al. ⁶⁸	302	35	0	Canada	individualist	clinical anger	non-offenders	State	IAEC/State ANG	Suppression	STAXI/ANG IN	0.34
Breen and Kashdan ⁶⁹	170	21.29	75.3	USA	individualist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Avoidance	AAQ-II	0.37
Breen and Kashdan ⁶⁹	170	21.29	75.3	USA	individualist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Suppression	STAXI-2/ANG IN	0.32
Burns, et al. ⁷⁰	105	46.3	48.6	USA	individualist	non-clinic	non-offenders	State	Anger 3-item 9-point Likert	Suppression	Suppression 4-item 9-point Likert	0.38
Carr ⁷¹ (Time 1)	164	69.15	73.17	USA	individualist	non-clinic	non-offenders	Trait	Anger 3-item	Reappraisal	Strategies for coping with widow(er)hood/Positive reframing	-0.05
Carr ⁷¹ (Time 2)	164	69.15	73.17	USA	individualist	non-clinic	non-offenders	Trait	Anger 3-item	Reappraisal	Strategies for coping with widow(er)hood/Positive reframing	0.03
Cheung and Park ⁷²	365	19.3	55.9	USA	individualist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Suppression	STAXI-2/ANG IN	0.35
Cho, et al. ⁷³	605	41.33	55.2	USA	individualist	non-clinic	non-offenders	Event-related State	Anger 3-item 7-point Likert	Acceptance	Acceptance 3-item 7-point Likert	0.04
Cho, et al. ⁷³	605	41.33	55.2	USA	individualist	non-clinic	non-offenders	Event-related State	Anger 3-item 7-point Likert	Avoidance	Avoidance 3-item 7-point Likert	-0.10
Cludius, et al. ⁷⁴	48	32.46	50	Germany	individualist	clinic	non-offenders	Trait	STAXI-2/Trait ANG	Acceptance	DERS/Nonacceptance	-0.59
Cludius, et al. ⁷⁴	48	32.46	50	Germany	individualist	clinic	non-offenders	Trait	STAXI-2/Trait ANG	Suppression	STAXI-2/ANG IN	0.46
Coleman, et al. ⁷⁵ (Time 1)	96	32.4	0	USA	individualist	non-clinic	non-offenders	Event-related State	Anger 3-item 6-point Likert	Rumination	9-item DRS	0.47

Continued

Study reference	N	Mean age	% women	Country	Individualist/Collectivist	Clinical status	Criminal status	Type of anger	Anger scale	Type of ER strategy	ER strategy scale	Effect size (r)
Coleman, et al. ⁷⁵ (Time 2)	80	32.4	0	USA	individualist	non-clinic	non-offenders	Event-related State	Anger 3-item 6-point Likert	Rumination	9-item DRS	0.61
Conrad, et al. ⁷⁶ (Sample 1)	94	35.75	62.8	Germany	individualist	non-clinic	non-offenders	Trait	STAXI/Trait ANG	Suppression	STAXI/ANG IN	0.24
Conrad, et al. ⁷⁶ (Sample 2)	321	38.83	58.9	Germany	individualist	clinic	non-offenders	Trait	STAXI/Trait ANG	Suppression	STAXI/ANG IN	0.39
Consedine, et al. ⁷⁷	1361	59.3	100	USA	individualist	non-clinic	non-offenders	Trait	STAXI/Trait ANG	Suppression	PPQ	0.24
Crisan and Nechita ⁷⁸	203	25.55	84.7	Romania	collectivist	clinic	non-offenders	Trait	STAXI-2/Trait ANG	Avoidance	AAQ-II	0.16
Crisan and Nechita ⁷⁸	203	25.55	84.7	Romania	collectivist	clinic	non-offenders	Trait	STAXI-2/Trait ANG	Rumination	ARS	0.45
da Costa, et al. ⁷⁹	98	31.3	28.6	Portugal	collectivist	non-clinic	non-offenders	State	STAXI/State ANG	Reappraisal	ERQ/R	0.07
da Costa, et al. ⁷⁹	98	31.3	28.6	Portugal	collectivist	non-clinic	non-offenders	Trait	STAXI/Trait ANG-Temp	Reappraisal	ERQ/R	0.01
da Costa, et al. ⁷⁹	98	31.3	28.6	Portugal	collectivist	non-clinic	non-offenders	State	STAXI/State ANG	Suppression	ERQ/S	0.04
da Costa, et al. ⁷⁹	98	31.3	28.6	Portugal	collectivist	non-clinic	non-offenders	Trait	STAXI/Trait ANG-Temp	Suppression	ERQ/S	-0.03
Dahlen and Deffenbacher ⁸⁰	86	19.3	76.74	USA	individualist	clinical anger	non-offenders	Event-related State	Anger 7-day log	Suppression	STAXI/ANG IN	0.08
Dahlen and Deffenbacher ⁸⁰	86	19.3	76.74	USA	individualist	clinical anger	non-offenders	Trait	STAXI/Trait ANG	Suppression	STAXI/ANG IN	0.04
Deffenbacher, et al. ⁸¹	48	-	45.83	USA	individualist	clinical anger	non-offenders	Event-related State	STAXI/State ANG	Suppression	STAXI/ANG IN	0.17
Deffenbacher, et al. ⁸¹	48	-	45.83	USA	individualist	clinical anger	non-offenders	Trait	STAXI/Trait ANG	Suppression	STAXI/ANG IN	0.42
Dennison and Stewart ⁸²	222	25.5	76	Australia	individualist	non-clinic	non-offenders	Trait	List of descriptors for Anger/1-factor Anger	Rumination	List of descriptors/1-factor Rumination	0.14
Donahue, et al. ⁸³	23	54.83	0	USA	individualist	clinical anger	non-offenders	Trait	Anger DAR	Avoidance	AAQ-II	0.79
Emmerich, et al. ⁸⁴	397	45.83	90.5	German countries	individualist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Suppression	STAXI-2/ANG IN	0.20
Franco, et al. ⁸⁵	36	41.27	100	Spain	individualist	non-clinic	non-offenders	Trait	POMS/Anger	Avoidance	AAQ	0.14
Fresnes and Borders ⁸⁶	201	19.48	78	USA	individualist	non-clinic	non-offenders	State	STAXI-2/State ANG	Rumination	ARS	0.33
Garofalo, et al. ⁸⁷ (Sample 1)	197	38.88	0	Italy	individualist	non-clinic	non-offenders	Trait	BPACQ/Anger	Acceptance	DERS/Nonacceptance	-0.33
Garofalo, et al. ⁸⁷ (Sample 2)	153	41.78	0	Italy	individualist	non-clinic	offenders	Trait	BPACQ/Anger	Acceptance	DERS/Nonacceptance	-0.36
Garofalo, et al. ⁸⁸ (Sample 1)	245	38.9	0	Italy	individualist	non-clinic	non-offenders	Trait	BPACQ/Anger	Acceptance	DERS/Nonacceptance	-0.39
Garofalo, et al. ⁸⁸ (Sample 2)	221	40.9	0	Italy	individualist	non-clinic	offenders	Trait	BPACQ/Anger	Acceptance	DERS/Nonacceptance	-0.44
Garofalo, et al. ⁸⁹ (Sample 1)	324	37.87	0	Italy	individualist	non-clinic	non-offenders	Trait	BPACQ/Anger	Acceptance	DERS/Nonacceptance	-0.38
Garofalo, et al. ⁸⁹ (Sample 2)	397	40.26	0	Italy	individualist	non-clinic	offenders	Trait	BPACQ/Anger	Acceptance	DERS/Nonacceptance	-0.38
Halperin and Gross ⁹⁰	201	-	49.75	Israel	individualist	non-clinic	non-offenders	Event-related State	Anger 1-item 6-point Likert	Reappraisal	3-item abbreviated version of ERQ/R	0.08
Han, et al. ⁹¹	371	-	80.32	China	collectivist	non-clinic	non-offenders	Trait	NAS/part B	Rumination	RRS	0.29

Continued

Study reference	N	Mean age	% women	Country	Individualist/Collectivist	Clinical status	Criminal status	Type of anger	Anger scale	Type of ER strategy	ER strategy scale	Effect size (<i>r</i>)
Haukkala ⁹² (Sample 1)	1402	46	0	Finland	individualist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Suppression	STAXI-2/ANG IN	0.30
Haukkala ⁹² (Sample 2)	1642	45.5	100	Finland	individualist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Suppression	STAXI-2/ANG IN	0.22
Jordaan and Hesselink ⁹³	243	25	0	South Africa	individualist	non-clinic	offenders	Trait	BPAQ/Anger	Avoidance	CSI/Avoidance	0.26
Kale and Gedik ⁹⁴	156	27.25	10.3	Turkey	collectivist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Reappraisal	ERQ-R	-0.14
Kale and Gedik ⁹⁴	156	27.25	10.3	Turkey	collectivist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Suppression	STAXI-2/ANG IN	0.51
Kitayama, et al. ⁹⁵ (Sample 1)	1050	58.04	55.04	USA	individualist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Suppression	STAXI-2/ANG IN	0.49
Kitayama, et al. ⁹⁵ (Sample 2)	362	55.47	59.11	Japan	collectivist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Suppression	STAXI-2/ANG IN	0.43
Kong, et al. ⁹⁶	573	22.26	49.56	China	collectivist	non-clinic	non-offenders	Trait	STAXI-2/ANG	Rumination	Rumination 7-item 5-point Likert	0.36
Kosson, et al. ⁹⁷ (Sample 1)	122	40.17	53.5	Italy	individualist	non-clinic	non-offenders	Trait	DES-IV Trait Anger	Suppression	STAXI-2/ANG IN	0.32
Kosson, et al. ⁹⁷ (Sample 2)	161	41.49	0	Italy	individualist	non-clinic	offenders	Trait	DES-IV Trait Anger	Suppression	STAXI-2/ANG IN	0.53
Lee, et al. ⁹⁸	189	43.01	56.6	Germany	individualist	non-clinic	non-offenders	Trait	AEQ-Teacher/ANG	Reappraisal	ERQ/R	0.04
Lee, et al. ⁹⁸	189	43.01	56.6	Germany	individualist	non-clinic	non-offenders	Trait	AEQ-Teacher/ANG	Suppression	ERQ/S	0.05
Li and Xia ⁹⁹ (Time 1)	1209	18.92	62	China	collectivist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Rumination	DAQ/Anger Rumination	0.50
Li and Xia ⁹⁹ (Time 2)	1209	18.92	62	China	collectivist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Rumination	DAQ/Anger Rumination	0.49
Li and Xia ⁹⁹ (Time 3)	1209	18.92	62	China	collectivist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Rumination	DAQ/Anger Rumination	0.44
Li, et al. ¹⁰⁰	958	-	62.7	China	collectivist	non-clinic	non-offenders	Event-related State	Anger 3-item 5-point Likert	Reappraisal	Reappraisal 4-item 5-point Likert	-0.12
Li, et al. ¹⁰⁰	958	-	62.7	China	collectivist	non-clinic	non-offenders	Event-related State	Anger 3-item 5-point Likert	Suppression	Suppression 2-item 5-point Likert	0.06
Martin and Dahlen ³⁶	362	20.46	79	USA	individualist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Reappraisal	CERQ/Positive Reappraisal	-0.22
Martin and Dahlen ³⁶	362	20.46	79	USA	individualist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Rumination	CERQ/Rumination	0.26
Martin and Dahlen ³⁶	362	20.46	79	USA	individualist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Suppression	STAXI-2/ANG IN	0.29
Massa, et al. ¹⁰¹	271	33.14	36.6	USA	individualist	non-clinic	offenders	Trait	BPAQ/Anger	Rumination	RRQ	0.50
Mathes, et al. ¹⁰²	119	42.26	48.7	USA	individualist	clinic	non-offenders	Trait	BPAQ/Anger	Rumination	RRS	0.45
McGonigal and Dixon-Gordon ¹⁰³	30	29.87	0	USA	individualist	non-clinic	offenders	Trait	STAXI-2/Trait ANG	Acceptance	DERS/Nonacceptance	-0.46
Monaci and Veronesi ¹⁰⁴ (Sample 1)	88	-	100	Italy	individualist	non-clinic	non-offenders	Event-related State	STAXI-2/State ANG	Suppression	STAXI-2/ANG IN	0.34
Monaci and Veronesi ¹⁰⁴ (Sample 2)	92	-	0	Italy	individualist	non-clinic	non-offenders	Event-related State	STAXI-2/State ANG	Suppression	STAXI-2/ANG IN	0.19

Continued

Study reference	N	Mean age	% women	Country	Individualist/Collectivist	Clinical status	Criminal status	Type of anger	Anger scale	Type of ER strategy	ER strategy scale	Effect size (<i>r</i>)
Moron and Biolik-Moron ¹⁰⁵ (Study 1)	218	29.6	45.87	Poland	individualist	non-clinic	non-offenders	Trait	BPAQ/Anger	Suppression	CECS/Anger suppression	-0.17
Moron and Biolik-Moron ¹⁰⁵ (Study 2)	103	27.5	69.9	Poland	individualist	non-clinic	non-offenders	Event-related State	STAXI-2/State ANG	Avoidance	AAQ-II	0.08
Moron and Biolik-Moron ¹⁰⁵ (Study 2)	103	27.5	69.9	Poland	individualist	non-clinic	non-offenders	Event-related State	STAXI-2/State ANG	Rumination	DAQ/Anger Rumination-4 items selected	0.39
Moron and Biolik-Moron ¹⁰⁵ (Study 2)	103	27.5	69.9	Poland	individualist	non-clinic	non-offenders	Event-related State	STAXI-2/State ANG	Suppression	STAXI-2/ANG IN	0.18
Navarro-Haro, et al. ¹⁰⁶	109	27.38	100	Spain	individualist	clinic	non-offenders	State	STAXI-2/State ANG	Reappraisal	ERQ/R	-0.15
Navarro-Haro, et al. ¹⁰⁶	109	27.38	100	Spain	individualist	clinic	non-offenders	Trait	STAXI-2/Trait ANG	Reappraisal	ERQ/R	-0.41
Navarro-Haro, et al. ¹⁰⁶	109	27.38	100	Spain	individualist	clinic	non-offenders	State	STAXI-2/State ANG	Suppression	ERQ/S	-0.12
Navarro-Haro, et al. ¹⁰⁶	109	27.38	100	Spain	individualist	clinic	non-offenders	Trait	STAXI-2/Trait ANG	Suppression	ERQ/S	-0.14
Offredi, et al. ¹⁰⁷	45	22.79	66.66	Italy	individualist	non-clinic	non-offenders	Trait	STAXI/Trait ANG	Rumination	ARS	0.28
Park, et al. ¹⁰⁸	233	53.59	67.38	USA	individualist	non-clinic	non-offenders	Trait	Anger 1-item 5-point Likert	Suppression	STAXI/ANG IN	-0.14
Peled and Moretti ⁵¹	226	19.7	68.58	USA	individualist	non-clinic	non-offenders	Trait	STAXI/Trait ANG- Temp	Rumination	SARI	0.33
Peters, et al. ¹⁰⁹	823	19.25	70.6	USA	individualist	non-clinic	non-offenders	Trait	BPAQ/Anger	Rumination	ARS	0.44
Prikhidko and Swank ¹¹⁰	227	-	100	USA	individualist	non-clinic	non-offenders	Trait	PAS/Anger experience	Reappraisal	Revised ERQ-9	-0.27
Prikhidko and Swank ¹¹⁰	227	-	100	USA	individualist	non-clinic	non-offenders	Trait	PAS/Anger experience	Suppression	Revised ERQ-9	0.00
Quan, et al. ¹¹¹	600	20.51	51.67	China	collectivist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Rumination	ARS	0.49
Salguero et al. ¹¹² (Study 1)	314	35.49	80.6	Australia	individualist	non-clinic	non-offenders	Trait	DAR-R	Rumination	ARS	0.61
Salguero et al. ¹¹² (Study 2)	633	36.62	56.9	Spain	individualist	non-clinic	non-offenders	Trait	DAR-R	Rumination	ARS	0.57
Sander, et al. ¹¹³	165	43.7	94	USA	individualist	non-clinic	non-offenders	Trait	PROMIS Anger	Suppression	TBI-CareQOL Emotional Suppression	0.29
Schröder-Abé, et al. ¹¹⁴	79	23.73	69.62	Germany	individualist	non-clinic	non-offenders	Trait	STAXI/Trait ANG	Suppression	STAXI/ANG IN	0.08
Shallcross, et al. ¹¹⁵ (Time 1)	340	41.32	48	USA	individualist	non-clinic	non-offenders	Trait	Anger 2-item 5-point Likert	Acceptance	KIMS/ Acceptance	-0.29
Shallcross, et al. ¹¹⁵ (Time 2)	280	41.32	48	USA	individualist	non-clinic	non-offenders	Event-related State	Anger 2-item 5-point Likert	Acceptance	KIMS/ Acceptance	-0.18
Shallcross, et al. ¹¹⁵ (Time 3)	244	41.32	48	USA	individualist	non-clinic	non-offenders	State	Anger 2-item 5-point Likert	Acceptance	KIMS/ Acceptance	-0.25
Shallcross, et al. ¹¹⁵ (Time 4)	239	41.32	48	USA	individualist	non-clinic	non-offenders	Trait	Anger 2-item 5-point Likert	Acceptance	KIMS/ Acceptance	-0.15
Continued												

Study reference	N	Mean age	% women	Country	Individualist/Collectivist	Clinical status	Criminal status	Type of anger	Anger scale	Type of ER strategy	ER strategy scale	Effect size (<i>r</i>)
Smeijers, et al. ¹¹⁶	413	20.04	82.6	the Netherlands	individualist	non-clinic	non-offenders	State	STAXI/State ANG	Rumination	DAQ/Anger rumination	0.31
Smeijers, et al. ¹¹⁶	413	20.04	82.6	the Netherlands	individualist	non-clinic	non-offenders	Trait	STAXI/Trait ANG	Rumination	DAQ/Anger rumination	0.47
Sofia and Cruz ¹¹⁷	269	21.73	28	Portugal	collectivist	non-clinic	non-offenders	Trait	CAAS/Anger	Reappraisal	ERSCS/ Reappraisal	-0.13
Sofia and Cruz ¹¹⁷	269	21.73	28	Portugal	collectivist	non-clinic	non-offenders	Trait	CAAS/Anger	Rumination	DAQ/Anger rumination	0.31
Sofia and Cruz ¹¹⁷	269	21.73	28	Portugal	collectivist	non-clinic	non-offenders	Trait	CAAS/Anger	Suppression	'White Bear' Suppression Inventory	0.20
Solak, et al. ¹¹⁸	131	21.6	67.17	Turkey	collectivist	non-clinic	non-offenders	Trait	Anger 3-item 7-point Likert	Suppression	ERQ/S	-0.08
Stover, et al. ¹¹⁹	378	33.59	0	USA	individualist	non-clinic	offenders	Trait	MAI/Anger	Acceptance	DERS/Nonacceptance	-0.35
Sullivan and Kahn ¹²⁰	97	20.59	53.6	USA	individualist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Suppression	ERQ/S	0.13
Sun, et al. ¹²¹	3025	-	63.3	China	collectivist	non-clinic	non-offenders	State	Anger 1-item 5-option	Reappraisal	ERQ/R	-0.14
Sun, et al. ¹²¹	3025	-	63.3	China	collectivist	non-clinic	non-offenders	State	Anger 1-item 5-option	Suppression	ERQ/S	-0.06
Thomsen, et al. ¹²²	126	23.32	60.31	Denmark	individualist	non-clinic	non-offenders	Trait	POMS-37/ANG	Rumination	ECQ-R	0.23
Toohey, et al. ¹²³	79	52.1	7.59	USA	individualist	clinic	non-offenders	State	STAXI-2/State ANG	Avoidance	AAQ-II	0.22
Toohey, et al. ¹²³	79	52.1	7.59	USA	individualist	clinic	non-offenders	Trait	STAXI-2/Trait ANG	Avoidance	AAQ-II	0.12
Torres-Marin, et al. ¹²⁴	164	28.8	53.04	Spain	individualist	non-clinic	non-offenders	Trait	STAXI/Trait ANG	Suppression	STAXI/ANG IN	0.26
Trew and Alden ¹²⁵	363	19.55	73.3	Canada	individualist	non-clinic	non-offenders	Trait	STAXI/Trait ANG	Rumination	RRS-Brooding	0.33
Trew and Alden ¹²⁵	363	19.55	73.3	Canada	individualist	non-clinic	non-offenders	Trait	STAXI/Trait ANG	Rumination	RRS-Reflective Pondering	0.14
Trew and Alden ¹²⁵	363	19.55	73.3	Canada	individualist	non-clinic	non-offenders	Trait	STAXI/Trait ANG	Suppression	STAXI/ANG IN	0.44
Tull, et al. ¹²⁶	113	26.41	0	USA	individualist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Avoidance	AAQ	0.19
van Middendorp, et al. ¹²⁷	331	47	100	the Netherlands	individualist	non-clinic	non-offenders	Event-related State	Anger 2 item 5-point Likert	Suppression	SECS/ANG IN	0.14
Velotti, et al. ¹²⁸	111	41.34	0	Italy & Australia	individualist	non-clinic	offenders	State	STAXI-2/State ANG	Acceptance	DERS/Nonacceptance	-0.27
Velotti, et al. ¹²⁸	111	41.34	0	Italy & Australia	individualist	non-clinic	offenders	Trait	STAXI-2/Trait ANG	Acceptance	DERS/Nonacceptance	-0.15
Velotti, et al. ¹²⁸	111	-	0	Italy & Australia	individualist	non-clinic	offenders	State	STAXI-2/State ANG	Suppression	STAXI/ANG IN	0.23
Velotti, et al. ¹²⁸	111	-	0	Italy & Australia	individualist	non-clinic	offenders	Trait	STAXI-2/Trait ANG	Suppression	STAXI/ANG IN	0.34
Wang, et al. ¹²⁹	464	20.75	65.1	China	collectivist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Rumination	ARS	0.70
Continued												

Study reference	N	Mean age	% women	Country	Individualist/Collectivist	Clinical status	Criminal status	Type of anger	Anger scale	Type of ER strategy	ER strategy scale	Effect size (<i>r</i>)
Weber, et al. ¹³⁰	93	23.6	68	Germany	individualist	non-clinic	non-offenders	Trait	STAXI/Trait ANG	Reappraisal	ERQ-R	-0.20
Weindl, et al. ¹³¹	220	57.9	40	Austria	individualist	non-clinic	non-offenders	Trait	STAXI-2/Trait ANG	Rumination	DAQ/Anger rumination	0.57
Wu, et al. ¹³²	358	19.18	63.12	China	collectivist	non-clinic	non-offenders	Trait	STAXI/Trait ANG	Rumination	RRQ-R	0.28
Yavuz, et al. ¹³³	34	31.6	0	Turkey	collectivist	clinic	non-offenders	Trait	STAXI/Trait ANG	Avoidance	AAQ-II	0.54

Table 1. Characteristics of studies included in the meta-analysis. AAQ-II, Acceptance and Action Questionnaire – version 2; AEQ, Achievement Emotions Questionnaire; ANG, anger; ANG IN, Anger In; BPAQ, Buss Perry Aggression Questionnaire; CAAS, Competitive Anger and Aggressiveness Scale; CECs, Courtauld Emotional Control Scale; CERQ, Cognitive Emotion Regulation Questionnaire; CoSS, Compass of Shame Scale; CWSS, The Coping with Stress Scale; DAR, Dimensions of Anger Reactions; DAQ, Displaced Aggression Questionnaire; DEI, Discrete Emotions Inventory; DERS, Difficulties in Emotion Regulation Scale; DES-IV, Differential Emotions Scale version IV; DRS, Dissipation-Rumination Scale; ECQ-R, Emotion Control Questionnaire – Rumination subscale; ERQ, Emotion Regulation Questionnaire (R, Reappraisal; S, Suppression); ERSCS, Emotion Regulation During Sport Competition Scale; KIMS, Kentucky Inventory of Mindfulness Scale; MAI, Multidimensional Anger Test; ML-STAXI, State Trait Anger Expression and Control Inventory; NAS, Novaco Anger Scale; PAS, Parent Anger Scale; POMS, Profile of Mood Scale; PROMIS, Patient-Reported Outcomes Measurement Information System; PPQ, Present Personality Questionnaire; RIOS, Rumination about Interpersonal Offense Scale; RRQ, Rumination-Reflection Questionnaire; RRS, Rumination Response Scale; SARI, Sadness and Anger Rumination Inventory; SECS, Self-Expression and Control Scale; STAXI-2, State-Trait Anger Expression Inventory, 2nd Edition; TBI-CareQOL, Traumatic Brain Injury Caregiver Quality of Life; Tehran MAS, The Tehran Multidimensional Anger Scale

Strategy	<i>k</i> (# effect sizes)	<i>r</i>	95% CI	<i>Z</i>	<i>p</i> (<i>Z</i>)	<i>Q</i> (<i>df</i>)	<i>p</i> (<i>Q</i>)	<i>I</i> ²
Acceptance	9 (12)	-0.32	-0.41; -0.23	-6.49	<0.001	101.03(11)	<0.001	89.11
Avoidance	12 (12)	0.19	0.08; 0.30	3.43	0.001	78.88(11)	<0.001	86.05
Distraction	1 (1)	-	-	-	-	-	-	-
Reappraisal	12 (12)	-0.13	-0.17; -0.05	-3.98	<0.001	38.82(11)	<0.001	71.67
Rumination	32 (37)	0.42	0.38; 0.47	16.23	<0.001	351.82(36)	<0.001	89.76
Suppression	35 (41)	0.24	0.17; 0.30	6.89	<0.001	792.56(40)	<0.001	94.95

Table 2. Meta-analytic associations between anger and emotion regulation strategies. CI, confidence interval; df, degrees of freedom.

Publication bias

The funnel plots (Fig. 2A-E) suggested the distribution of the effect sizes was symmetrical for the relations between anger and reappraisal, rumination, and suppression. For the relations of anger with both acceptance and avoidance, the funnel plots showed some asymmetry to the left.

The trim-and-fill procedure imputed two studies in the analysis of the relation between anger and acceptance, which slightly decreased the effect size ($r = -0.27$). One study was imputed in analysis on anger and reappraisal, and the effect size decreased slightly ($r = -0.11$). It also imputed three studies in the analysis of the relation between anger and suppression, which decreased the effect size ($r = 0.19$). No study was imputed in the analysis on anger and avoidance, and anger and rumination.

Egger’s test also indicated that the distribution of effect sizes was symmetrical in all analyses (all $ps \geq 0.069$), except that on the relation between anger and avoidance ($p = 0.007$).

Discussion

The present results supported consistent associations between anger and multiple emotion regulation strategies, ranging from small to large effect sizes. In addition, the present results identified multiple study-level differences that explained heterogeneity in previous results. They also highlighted multiple methodological limitations of previous studies and suggest critical issues that should be considered in future work.

Anger was positively associated with avoidance, rumination and suppression. These results are in line with theories which have argued that problematic anger involves exaggerated efforts to avoid or block this emotion^{14,15}. They also accord with experimental evidence showing that rumination and suppression increase and prolong anger^{25,27}. Anger was also negatively associated with the use of acceptance and reappraisal. Theories of psychopathology view the former strategy as fundamental for adaptive emotion regulation²⁸, whereas the latter strategy plays a central role in cognitive-behavioral interventions¹⁸.

Are the present results trivial or do they offer a much-needed perspective on a large and heterogeneous literature? Despite apparent theoretical agreement, it is noteworthy that previous evidence has not always supported the expected pattern of associations between anger and emotion regulation strategies. For instance, while most studies reported positive associations between anger and both avoidance and suppression, and negative associations between anger and both acceptance and reappraisal, some of the previous studies reported correlations in the opposite direction. Furthermore, the effect sizes of these correlations—including rumination—varied widely, ranging from small to large. Therefore, we believe that the present meta-analysis was necessary and contributes to the field by examining whether the previously reported associations between anger and emotion regulation strategies are consistent across studies.

Another question is whether the pattern of emotion regulation strategies associated with anger in the present meta-analysis converges with the results of previous meta-analytic efforts across emotions. Evidence suggests that emotion regulation may partially differ between emotions. For instance, a study found that in anger compared to sadness, people used more emotion regulation strategies, but the strategies did not completely overlap and their effectiveness also differed¹³⁴. Avoidance was used in anger more than in sadness, and the reverse was true for reappraisal¹³⁴. Furthermore, avoidance was associated with lower regulation effectiveness in anger, but not in sadness¹³⁴. A meta-analysis also supported the view that the efficiency of emotion regulation differed between emotions, being lowest in anger and highest in sadness, among the negative emotions that were analyzed³⁸. However, this comparison between emotions was possible only across strategies. The present results extend this perspective by identifying the emotion regulation strategies that are positively (i.e., avoidance, rumination, suppression) and negatively (i.e., acceptance, reappraisal) associated with anger. This pattern seems partially different from the one found across emotions³⁸ in that suppression is positively associated with anger (and negatively across emotions), and the effect sizes are larger for anger than across emotions.

Another contribution of the present meta-analysis relates to the identification of study characteristics that were found to consistently influence the relations between anger and emotion regulation. For instance, the association between anger and suppression was stronger in studies assessing trait anger compared to those assessing state anger related to a specific event, while studies assessing state anger without a reference event found no significant association. This supports the view that by focusing on context-independent dimensions of anger, dispositional measures capture more of the association with suppression compared to state anger. An alternative explanation is that the larger effect size reflects the similar focus of both trait anger and habitual suppression measures on dispositional aspects (i.e., how one is in general). Assessing state anger without specifying or controlling for the reference event is not recommended because it may introduce uncontrolled contextual variance.

Outcome variable	Moderator variable		k	r	95% CI	Z	p(Z)	Q(df)	P(Q)
Avoidance	Type of anger	Trait anger	-	-	-	-	-	-	-
		State anger related to a specific event	-	-	-	-	-		
		State anger not related to a specific event	-	-	-	-	-		
	Clinical status	Clinical diagnosis	3	0.24	[0.04; 0.43]	2.39	0.017	24.32(1)	0.001
		Clinical anger	-	-	-	-	-		
		Non-clinical	8	0.12	[0.00; 0.23]	2.08	0.037		
	Criminal status	Offenders	-	-	-	-	-	-	-
		Non-offenders	-	-	-	-	-		
	Type of culture	Individualist	8	0.22	[0.07; 0.35]	2.87	0.004	0.05(1)	0.813
		Collectivist	4	0.16	[-0.04; 0.35]	1.58	0.113		
	Type of design	Cross-sectional	-	-	-	-	-	-	-
		Longitudinal	-	-	-	-	-		
	Study quality	Fair	4	0.25	[0.12; 0.37]	3.72	<0.001	12.93(1)	0.008
		Poor	8	0.16	[0.03; 0.29]	2.40	0.016		
Acceptance	Type of anger	Trait anger	-	-	-	-	-	-	-
		State anger related to a specific event	-	-	-	-	-		
		State anger not related to a specific event	-	-	-	-	-		
	Clinical status	Clinical diagnosis	-	-	-	-	-	-	-
		Clinical anger	-	-	-	-	-		
		Non-clinical	-	-	-	-	-		
	Criminal status	Offenders	6	-0.35	[-0.41; -0.29]	-10.06	<0.001	20.24(1)	<0.001
		Non-offenders	6	-0.30	[-0.44; -0.14]	-3.71	<0.001		
	Type of culture	Individualist	-	-	-	-	-	-	-
		Collectivist	-	-	-	-	-		
	Type of design	Cross-sectional	-	-	-	-	-	-	-
		Longitudinal	-	-	-	-	-		
	Study quality	Fair	5	-0.37	[-0.46; -0.27]	-7.04	<0.001	13.88(1)	<0.001
		Poor	7	-0.29	[-0.43; -0.13]	-3.52	<0.001		
Reappraisal	Type of anger	Trait anger	7	-0.13	[-0.21; -0.04]	-2.95	0.003	0.99(1)	0.607
		State anger related to a specific event	-	-	-	-	-		
		State anger not related to a specific event	3	-0.09	[-0.20; 0.01]	-1.728	0.084		
	Clinical status	Clinical diagnosis	-	-	-	-	-	-	-
		Clinical anger	-	-	-	-	-		
		Non-clinical	-	-	-	-	-		
	Criminal status	Offenders	-	-	-	-	-	-	-
		Non-offenders	-	-	-	-	-		
	Type of culture	Individualist	7	-0.13	[-0.22; -0.05]	-3.10	0.002	0.02(1)	0.883
		Collectivist	5	-0.12	[-0.23; -0.00]	-2.09	0.036		
	Type of design	Cross-sectional	-	-	-	-	-	-	-
		Longitudinal	-	-	-	-	-		
	Study quality	Fair	-	-	-	-	-	-	-
		Poor	-	-	-	-	-		
Rumination	Type of anger	Trait anger	28	0.42	[0.36; 0.47]	13.60	<0.001	2.96(2)	0.228
		State anger related to a specific event	6	0.44	[0.34; 0.53]	7.87	<0.001		
		State anger not related to a specific event	3	0.40	[0.21; 0.57]	3.86	<0.001		
	Clinical status	Clinical diagnosis	-	-	-	-	-	-	-
		Clinical anger	-	-	-	-	-		
		Non-clinical	-	-	-	-	-		
	Criminal status	Offenders	-	-	-	-	-	-	-
		Non-offenders	-	-	-	-	-		
	Type of culture	Individualist	28	0.41	[0.36; 0.46]	13.80	<0.001	30.87(1)	<0.001
		Collectivist	9	0.45	[0.36; 0.53]	9.04	<0.001		
	Type of design	Cross-sectional	-	-	-	-	-	-	-
		Longitudinal	-	-	-	-	-		
	Study quality	Fair	5	0.47	[0.37; 0.56]	8.40	<0.001	32.61(1)	<0.001
		Poor	31	0.42	[0.37; 0.47]	14.34	<0.001		
Continued									

Outcome variable	Moderator variable		k	r	95% CI	Z	p(Z)	Q(df)	P(Q)
Suppression	Type of anger	Trait anger	28	0.27	[0.20; 0.34]	7.76	< 0.001	297.52(2)	< .001
		State anger related to a specific event	7	0.13	[0.06; 0.20]	3.78	< 0.001		
		State anger not related to a specific event	6	0.13	[-0.05; 0.32]	1.38	0.165		
	Clinical status	Clinical diagnosis	3	0.21	[0.13; 0.29]	5.23	< 0.001	136.95(2)	< 0.001
		Clinical anger	5	0.45	[0.41; 0.49]	20.28	< 0.001		
		Non-clinical	33	0.18	[0.17; 0.20]	23.29	< 0.001		
	Criminal status	Offenders	-	-	-	-	-	-	-
		Non-offenders	-	-	-	-	-		
	Type of culture	Individualist	32	0.27	[0.25; 0.29]	29.55	< 0.001	106.29(1)	< 0.001
		Collectivist	9	0.11	[0.08; 0.13]	8.79	< 0.001		
	Type of design	Cross-sectional	-	-	-	-	-	-	-
		Longitudinal	-	-	-	-	-		
	Study quality	Fair	14	0.27	[0.18; 0.35]	5.95	< 0.001	41.16(1)	< 0.001
		Poor	27	0.22	[0.12; 0.31]	4.53	< 0.001		

Table 3. The results of subgroup analyses on potential moderators. *Note:* Line indicates that the analysis could not be run because there were insufficient studies ($k \geq 3$ in at least two subgroups). CI, confidence interval; df, degrees of freedom.

Outcome variable	Predictor variable	k	slope β	95% CI	p
Avoidance	Sample sex distribution (% women)	12	-0.00	[-0.00; 0.00]	0.096
	Sample mean age	12	0.00	[-0.00; 0.01]	0.348
Acceptance	Sample sex distribution (% women)	12	0.00	[0.00; 0.00]	0.028
	Sample mean age	12	0.02	[-0.00; 0.05]	0.054
Reappraisal	Sample sex distribution (% women)	11	0.00	[-0.00; 0.00]	0.093
	Sample mean age	8	0.00	[-0.00; 0.00]	0.068
Rumination	Sample sex distribution (% women)	33	-0.00	[-0.00; 0.00]	0.360
	Sample mean age	33	0.00	[-0.00; 0.01]	0.195
Suppression	Sample sex distribution (% women)	34	-0.00	[-0.00; -0.00]	0.052
	Sample mean age	34	0.00	[-0.00; 0.00]	0.783

Table 4. Results of meta-regression analyses on the relation of sex distribution and mean age with the effect size of the associations between anger and emotion regulation. CI, confidence interval.

As expected, we found that the positive relation between anger and both suppression and avoidance was larger in clinical compared to non-clinical samples. This is in line with the dimensional approach to psychopathology, which argues that clinical disorders are at the upper extreme of a continuum of individual differences or symptoms in the population¹³⁵. These results also support Robertson's model¹⁵, which has argued that one of the maladaptive styles underlying problematic anger involves emotion over-regulation, that is, an exaggerated use of suppression and avoidance aimed at blocking negative emotion. However, it is noteworthy that the association of anger with suppression was large in samples with clinically suggestive levels of anger (i.e., characterized by high levels of negative consequences), and medium in clinical samples selected for any diagnosis of a mental disorder. The likely explanation is that anger problems are present in multiple mental disorders, being more prominent in some disorders (e.g., oppositional defiant disorder, borderline personality disorder, antisocial personality disorder and bipolar disorders)^{136–139} than in others, and that focusing on individuals with high levels of this transdiagnostic symptom captures more of the association between anger and emotion regulation. One additional question is how many of the individuals with anger problems from previous samples had a mental disorder diagnosis? This is relevant in light of debates surrounding the development of specific anger disorder categories in clinical diagnostic⁴⁰. Unfortunately, only one out of five studies in clinical anger samples reported mental disorder diagnoses. This study⁸³ suggested a large overlap between anger problems and mental disorders, with 96% of the sample having been diagnosed with psychopathology. Future studies should report clinical status in order to allow for a more extensive examination of this issue.

The comparison between offenders and non-offenders revealed a stronger negative association between anger and acceptance in offenders. This is in line with the view that anger problems are associated with antisocial behaviors, and that low acceptance of anger may be fundamental in the maladaptive regulation of anger that underlies these behaviors^{14,15}. However, it is noteworthy that the effect size was also large in the non-offenders' group, which underscores the hypothesis that mechanisms other than acceptance may be involved in the relation between anger and antisocial behavior.

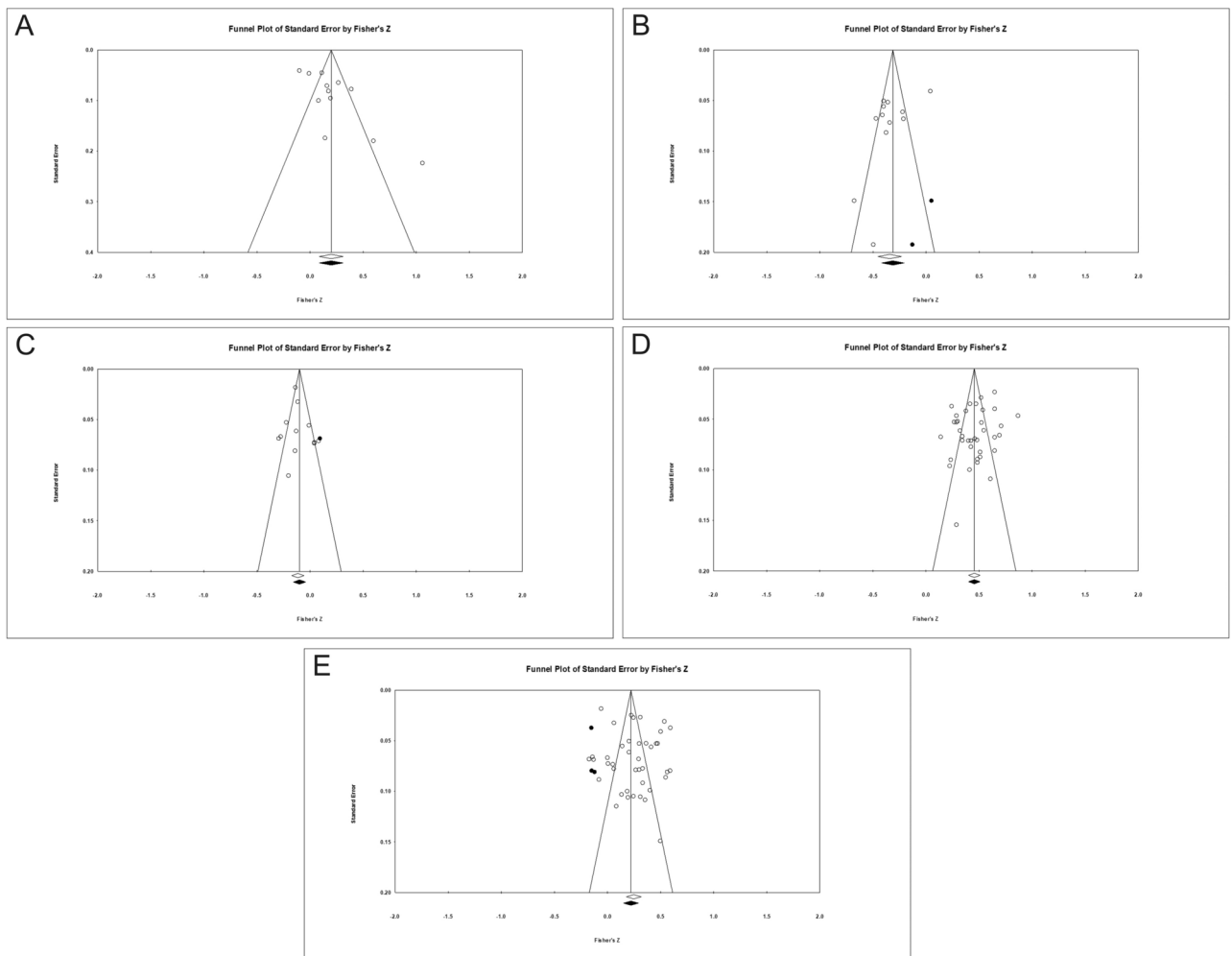


Fig. 2. Funnel plots of the effect sizes of the association of anger with (A) avoidance; (B) acceptance; (C) reappraisal; (D) rumination; and (E) suppression.

The positive association between anger and suppression was larger in samples from individualist compared to collectivist countries. As previously suggested⁴⁶, this may reflect that suppression, which typically involves holding back emotion for the sake of social relationships, is less congruent with the self-focused values promoted in individualist cultures. Therefore, when individuals from these cultures use suppression in situations in which someone is in the way of a personal goal, this may result in more anger. Conversely, perhaps by putting personal goals above social well-fare, individualist cultures could promote more anger, which in turn could entail more suppression given that this emotion can harm social image and social relationship. The present meta-analysis cannot discern whether culture promotes anger through suppression, or suppression through anger. The same applies to the positive association between anger and rumination, which was larger in samples from collectivist compared to individualist cultures.

The proportion of women in the sample was positively associated with the effect size of the relation between anger and acceptance. We see two potential explanations: compared to men, women may use acceptance to regulate anger more often, or they may use it more efficiently. Previous studies have generally not supported sex differences in acceptance across emotions^{140–142}, whereas the studies on anger acceptance included in the present meta-analysis did not examine sex differences. Therefore, we cannot provide a clear explanation, and it remains possible that this finding is driven by other sample characteristics that we did not control for. However, we caution researchers to consider sex differences in their future work on anger and emotion regulation, as these differences may otherwise confound their analyses.

Finally, for all emotion regulation strategies but suppression, the effect sizes were larger in the subgroup of studies rated as qualitatively better. This suggests that the present effects may have been underestimated due to the inclusion of studies that did not meet methodological rigors such as sample size justification, efforts to control potential confounds, and prospective design.

The present meta-analysis has several limitations, which reflect research gaps in the studies that were included. First, most of previous work is cross-sectional and correlational, which precludes any conclusion on the direction of the effects (i.e., anger drives emotion regulation strategies, vice-versa, or both ways). Given that longitudinal designs can shed light on the temporal relations between anger and emotion regulation, we

checked the results of the five studies included in the present meta-analysis. All focused on the prospective influence of anger on emotion regulation, and with one exception, did not investigate the reverse pathway. Li, & Xia³⁹ were the only ones to consider bidirectional pathways, and supported reciprocal effects between trait anger and rumination. We urge researchers to increasingly use longitudinal designs in future work and examine bidirectional pathways between anger and emotion regulation. This would also contribute to increasing study quality, which is considerably limited in previous studies, for reasons including the scarcity of longitudinal designs, but also issues related to justifying the sample size and controlling for confounding variables.

Second, there is only one study on anger and distraction⁷¹, and this study focused on behavioral rather than cognitive forms of distraction. Considering experimental evidence showing that when emotional intensity is high, people prefer distraction over reappraisal, and distraction is more efficient than reappraisal¹⁴³, future studies should investigate this strategy in anger. Furthermore, multiple subgroup analyses could not be run in the present analyses, which highlights specific aspects on which data is still limited.

Third, given the central role of emotion regulation in anger problems, studies should increasingly investigate this issue in clinical populations. Previous data is limited and did not allow us to compare the relations between anger and emotion regulation strategies between mental disorders. Anger problems are transdiagnostic, but they may be more prominent in certain disorders compared to others. Empirical investigations of this issue could crucially support efforts to define clinical anger.

Fourth, there was suggestive evidence that publication bias may have confounded the present analyses. While the results of statistical analyses of publication bias were not consistent, which is often the case and reflects the non-overlapping nature of these tests, all analyses except those on reappraisal and rumination showed some degree of publication bias.

Finally, we underscore the importance of focusing on the mediator role of emotion regulation in the relation between anger and aggressive behavior. There has been long-standing theoretical and clinical interest in this hypothesis, but data remains limited. Prospective correlational and experimental designs are the most appropriate approaches for investigating mediation and should be increasingly used in future work.

The present results support the view that anger is consistently associated with the differential use of multiple emotion regulation strategies. This offers an encouraging perspective on a long-standing hypothesis that has been at the heart of clinical theories and interventions in anger and suggests ways in which basic and clinical work in this area could be improved and extended.

Method

The current meta-analysis followed the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) statement¹⁴⁴ recommendations. The protocol was pre-registered in PROSPERO (registration number CRD42020214171).

Search strategy

The search was conducted in five bibliographical archives (PubMed; PsycINFO; Scopus; Web of Science; Cochrane Central Register of Controlled Trials), from inception until April 3, 2022. We searched for the following keywords in title and abstract: “anger”, “emotion regulation”, “reappraisal”, “suppression”, “distraction”, “rumination”, “acceptance”, “avoidance” (see Supplementary Materials for the complete search string). In addition, we searched for relevant articles in the reference section of previous reviews. The search was restricted to articles in English.

Eligibility criteria

Studies were eligible if they met the following criteria: (1) they were original quantitative studies, published in peer-reviewed journals; (2) they were conducted in humans; (3) they reported the relation between anger and one or more of the emotion regulation strategies (i.e., acceptance, reappraisal, avoidance, distraction, rumination, suppression); and (4) they provided sufficient statistical data for effect size estimation. We included both correlational studies and studies that compared groups with different levels of anger on emotion regulation.

The aim of this meta-analysis was to investigate the relationship between anger and the habitual or spontaneous use of emotion regulation strategies. Accordingly, we included studies that employed self-report questionnaires to assess the general use of these strategies, as well as field studies (e.g., ecological momentary assessment) that used self-report items to measure emotion regulation in specific contexts. Because these measures differ from those assessing the instructed use of emotion regulation strategies³⁹, we excluded studies in which emotion regulation strategies were manipulated. Studies were also excluded if they: (1) were conference abstracts, letters, dissertations or book chapters; (2) were reviews or meta-analyses not reporting original data; (3) were case studies or qualitative studies; and (4) used the same sample as another eligible study. Adults were the target population of the present meta-analysis and therefore, we excluded studies on samples with a mean age lower than 18 years.

Clarification of relevant constructs

Anger. We focused on anger, which has been distinguished from hostility, which refers to a pervasive aggressive attitude^{17,34,145,146}. In line with the phenomenological perspective¹⁴⁷, only measures of the subjective experience of anger were considered, and not measures of other aspects, such as physiological or behavioral responses related to anger. We included studies that focused on trait anger, state anger unrelated to a specific event, and state anger associated with a specific context (such as a real-world event or an anger-inducing manipulation, for which we considered only post-induction assessments).

Emotion regulation strategies. The following emotion regulation strategies were included in the analysis: (1) avoidance (i.e., reluctance to and refraining oneself from experiencing an emotional event); (2) acceptance

(i.e., allowing for emotional experiences to unfold naturally); (3) distraction (i.e., moving attention away from an emotional event); (4) cognitive reappraisal (i.e., changing one's perspective of an emotional event); (5) rumination (i.e., repetitive thinking about an emotional event or experience); and (6) suppression (i.e., inhibiting the behavioral expression of emotion).

Study selection

All retrieved articles were uploaded in EndNote, and duplicates were eliminated. The titles and abstracts were independently checked by two researchers in order to identify potentially relevant studies. The full texts of the selected studies were also independently assessed by two researchers, based on inclusion and exclusion criteria.

Data extraction and coding

For each study, the following data were extracted: (1) identification information (author, year); (2) sample information (i.e., sample size, mean age of the participants, percentage of female participants per study, clinical status, criminal status); (3) emotion regulation strategy; (4) type of anger (i.e., trait, state); (5) instrument used to assess anger; (6) instrument used to assess emotion regulation; (7) effect size; and (8) research design (i.e., cross-sectional, longitudinal). When the relevant data were not reported, authors were contacted for additional details.

The risk of bias in individual studies was assessed by two independent researchers, using tools provided by the National Heart, Lung, and Blood Institute. The Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies¹⁴⁸ was used for cross-sectional studies, and the Quality Assessment of Case-Control Studies¹⁴⁸ for studies in which individuals selected for high and low anger levels were compared. The methodological quality of studies was evaluated in line with the following characteristics: (1) clear research objective; (2) clear specification of the sample (eligibility criteria, justification for the sample size); (3) reliable and valid instruments; and (4) controlling of potential confounding variables. For each criterion, studies were categorized as “yes”, “no”, “cannot determine” (CD), “not applicable” (NA), or “not reported” (NR). In line with previous work^{149,150}, the quality of studies was considered good if more than 11 items were rated “yes”, fair if 6–9 items were rated “yes”, and poor if fewer than 6 items were rated “yes”.

Meta-analytic plan

Considering our focus on the association between anger and emotion regulation strategies, the Pearson correlation coefficient (r) was used as the index of effect size. In studies comparing individuals selected for anger, we used the group mean, standard deviation and sample size for independent groups to compute Pearson's r ¹⁵¹. Following the recommendations of Roth et al. (2018), we chose not to estimate the effect size based on beta coefficients. The analysis was conducted in Comprehensive Meta-analysis version 4.0. The effect sizes were first estimated as Fisher's z ¹⁵² and then transformed back into the original r -metric for interpretation and presentation¹⁵³. The magnitude of the effect sizes was interpreted in accordance with Gignac and Szodorai¹⁵⁴: small when $r = 0.10$, medium when $r = 0.20$, large when $r = 0.30$.

In light of the diversity of samples in the included studies, a random-effect model was thus employed in all analyses^{155,156}. In order to examine whether there is significant true variance (i.e., beyond random error) in effect sizes between studies, we assessed heterogeneity with the Q statistic test¹⁵⁷, and estimated the proportion of true variance using the I^2 . An I^2 index value of 0% indicates no observed heterogeneity, a value of 25% indicates low heterogeneity, of 50%, moderate heterogeneity, and of 75% or above, high heterogeneity¹⁵⁸.

To examine potential categorical moderators that might explain the true variance in effect sizes across studies, we performed subgroup analysis. The method involves dividing studies into subgroups based on a categorical moderator and comparing the effect sizes between subgroups using tests for heterogeneity (e.g., Cochran's Q) to determine if the differences are statistically significant. We examined the following categorical moderators: (1) type of anger: trait anger vs. state anger related to a specific event vs. state anger not related to a specific event; (2) clinical status: participants with a clinical diagnosis (i.e., any mental disorder diagnosed by a clinician using structured clinical interview) vs. participants with clinically-relevant anger problems (i.e., who scored in the upper quartile on the Trait Anger Scale, $TAS > 22$, and reported having severe anger problems) but no clinical disorder diagnosis, vs. nonclinical participants; (3) criminal status: offenders (i.e., participants who were imprisoned or perpetrated intimate partner violence, based on self- or other-reports) vs. non-offenders; (4) type of culture: collectivist vs. individualist, classified as such using the Hofstede's Country Comparison Tool (<https://www.hofstede-insights.com/country-comparison-tool>, accessed November 3, 2023); (5) study design: cross-sectional vs. longitudinal; and (6) study quality: good vs. fair vs. poor. In line with common practice in meta-analysis, a subgroup analysis was run only if there were at least three studies in at least two subgroups. For studies reporting effect sizes for multiple moderator categories on the same sample, in order to ensure that the assumption of independence was not violated in subgroup analyses, we selected only one effect size, specifically the one that added to the less represented subgroup.

Participants' demographic characteristics (i.e., percentage of women; mean age) and risk of bias across studies were considered potential continuous moderators. A restricted maximum likelihood model for the meta-regression analyses was performed to examine a potential relation between these quantitative study characteristics and the effect size.

Publication bias was examined in three ways: (1) visual inspection of the funnel plot; (b) the trim-and-fill method, which imputes missing studies that would render the funnel plot symmetrical, and estimates an adjusted effect size^{153,159}; and (c) the Egger test, which assesses whether the funnel plot is significantly asymmetrical¹⁶⁰.

Data availability

All data generated or analyzed in this study are included in this published article (and its Supplementary Material).

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Author contributions

GVP contributed to the conception and design of the work, the extraction, analysis and interpretation of data, and drafting the manuscript. DMN contributed to data extraction. ACM contributed to analysis and interpretation of data and drafting the manuscript. AST contributed to the conception and design of the work, analysis and interpretation of data, and revising the manuscript.

Declarations

Competing interests

One of the senior authors (Andrei C. Miu) is an Editorial Board Member, and one of the Guest Editors for the Emotion Regulation Collection. The other authors (GVP, DMN, AST) declare no competing interests.

Additional information

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